Python Program to Solve the 8-Puzzle Problem

Aim: To solve the 8-Puzzle problem using the A* search algorithm.

Python Code

```
from heapq import heappush, heappop

goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

def heuristic(state):
    return sum(state[i] != 0 and state[i] != goal[i]
               for i in range(9))

def solve(start):
    pq = []
    heappush(pq, (heuristic(start), 0, start, []))
    visited = set()

    while pq:
        f, g, state, path = heappop(pq)

        if state in visited:
            continue

        visited.add(state)
        path = path + [state]

        if state == goal:
            return path

        zero = state.index(0)
        moves = []

        if zero >= 3:
            moves.append(zero - 3)
        if zero < 6:
            moves.append(zero + 3)
        if zero % 3 != 0:
            moves.append(zero - 1)
        if zero % 3 != 2:
            moves.append(zero + 1)

        for pos in moves:
            new_state = list(state)
            new_state[zero], new_state[pos] = new_state[pos], new_state[zero]
            new_state = tuple(new_state)

            if new_state not in visited:
                heappush(pq, (g + 1 + heuristic(new_state),
                              g + 1, new_state, path))

    return None

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
result = solve(start)

if result:
    for state in result:
        print(state[0:3])
        print(state[3:6])
        print(state[6:9])
        print()
    print("Puzzle solved!")
else:
    print("No solution exists.")
```

Sample Output

```
(1, 2, 3)\n(4, 0, 6)\n(7, 5, 8)\n\n(1, 2, 3)\n(4, 5, 6)\n(7, 0, 8)\n\n(1, 2, 3)\n(4, 5, 6)\n(7, 8,
```

Algorithm

1. Initialize the starting state and goal state.
2. Calculate the heuristic using the number of misplaced tiles.
3. Select the state with the lowest estimated cost using A* search.
4. Move the blank tile (0) up, down, left, or right when the move is valid.

5. Repeat until the goal state is reached.
6. Print the solution path.

Result: The 8-Puzzle problem is solved successfully using the A* search algorithm.